

Hybrid effect on tensile properties of FRP rods with various material compositions

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Abstract

This paper deals with an experimental study on the tensile properties of hybrid FRP rods. The objective of this study is to identify a suitable material architecture for FRP reinforcing bars for concrete structures. Using carbon and E-glass fibers, three different material architectures were considered for hybrid FRP rods. Using vinylester and unsaturated polyester resins, the hybrid FRP rod samples were pultruded and tested for tension. The effect of hybridization on tensile properties of FRP rods was obtained by comparing the results of tensile test for the hybrid rods with those of the non-hybrid rods. The results of the test program indicated that the ultimate strain of the hybrid rod with dispersed type was increased by up to 33%.

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1. Introduction

The corrosion of steel reinforcing bar (rebar), as one of the main causes for the shortening of the service life of reinforced concrete (RC) structures, is the most cost-demanding feature for maintenance during the service life. The corrosion of steel bars is a material problem rather than structural one. To overcome this problem in RC structures, steel reinforcement should be coated or replaced with non-corrosive materials.

In a certain circumstance, Fiber Reinforced Polymer (FRP) is proposed as an alternative material to replace the structural steel in construction. If FRP is utilized as the internal reinforcement for concrete structures, it can provide high specific strength as well as good resistance to corrosion compared to the steel reinforcement [1].

Currently, many researches are actively applying FRP as the reinforcement or the structural materials in civil engi-

neering structures. In Korea, a research project is also being conducted to develop the FRP internal reinforcement for concrete structures [2].

Since FRP rebar generally exhibits higher tensile strength than the steel rebar with the same weight, a cross-section of concrete structures can be designed to support larger loads when the same ratio of reinforcement is used. Moreover, good resistance of FRP to corrosion may be an ideal material characteristic for the internal reinforcement used in concrete structures exposed to aggressive environments such as sea water.

FRP is mainly composed of fiber and resin. The commonly used fibers are glass and carbon. Carbon fiber exhibits higher tensile strength and more elastic modulus than steel. These are more advantageous features of carbon fiber in a structural point of view. Since its price is almost ten times as high as glass fiber, glass fiber is more beneficial material in initial cost. However, its low elastic modulus is a main disadvantage of glass fiber, which attains more less than a quarter of steel, and thus lead to excessive deflection when used as the reinforcement for flexural members. Since

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each fiber has strength as well as weakness, the concept of ‘hybridization’ should be employed for the FRP rebar to overcome each shortcoming.

This paper deals with an experimental study on the tensile properties of hybrid FRP rods that consist of carbon and E-glass fibers. The purpose of this study is to identify a suitable material architecture of the FRP reinforcing bar used for concrete structures. Three different material architectures, mainly the arrangement of fibers within the cross-section of FRP rod, were considered for the hybrid FRP rods. These arrangements of fibers are (a) concentrated in the core, (b) disposed at the outer surface (crust), and (c) dispersed over the cross-section.

Since the bond performance between the FRP rebar and concrete was beyond the scope of this study, FRP rebar with a circular cross-section was considered. Both vinyl-ester and unsaturated polyester were utilized as resin materials. FRP rods consisting of glass or carbon fibers were also fabricated for comparison. To examine the tensile properties of the hybrid and non-hybrid FRP rods, samples were pultruded and tested for tension. The effect of hybridization on tensile properties of FRP rods was obtained by comparing the results of tensile test for the hybrid rods with those of the non-hybrid rods.

2. Literature review

‘Hybrid composite’, which fiber composite members are called hereafter, is composed of more than two types of fiber. The purpose of hybridization is to create a new material that picks out only the advantages of each constituent. In general, ‘hybrid effect’ is the positive effect of any property obtained through the rule of mixture [3]. For instance, if the ultimate strain of an axial FRP member composed of more than two types of fiber is higher than that of a fiber with the lowest elongation, a positive hybrid effect is obtained by hybridization.

In general, the elastic modulus of a hybrid composite (E_H) reinforced in one direction can be obtained through the rule of mixture as follows:

$$E_H = \sum_{i=1}^n E_i V_i \quad \text{where} \quad \sum_{i=1}^n V_i = 1 \quad (1)$$

in which, E_i and V_i refer to the elastic modulus and the volume fraction of the i th constituent comprising the composite member.

If the induced strains of the various constituents within an axial composite member are uniform, the tensile strength can be calculated as the ultimate strain of the constituent with the lowest elongation. When a fracture of the fiber with the lowest elongation occurs, the stress induced in the hybrid composite (σ_H^{ult}) can be computed as,

$$\sigma_H^{\text{ult}} = \epsilon_f^{\text{ult}} E_H \quad (2)$$

where, ϵ_f^{ult} is the ultimate strain of the lowest elongation fiber and can be obtained from a fragmentation test for impregnated single fiber.

Kretsis [3] stated that a hybrid effect is more distinctive as the ratio and the concentration of low elongation fiber gets smaller. In 1994, Jones and DiBenedetto [4] carried out experiments on hybrid composite specimens constituted by carbon/carbon, glass/glass, aramid/carbon, and glass/carbon fibers. They revealed that pseudo-ductile behavior occurred to the specimen in which low elongation fiber was well distributed in small quantities, and Kretsis [3] confirmed the results.

In general, the fibers remaining after the failure of low elongation fibers can sustain additional applied load due to the stress redistribution. In 1976, Phillips [5] demonstrated that if these fibers do not fail immediately, the stress–strain relationship exhibits the pseudo-ductile behavior and can sustain the additional loads. On the basis of experimental results performed on the pultruded hybrid rods with different volume fraction and the arrangements of fibers (glass, aramid, carbon), Bakis et al. [6] reported that the specimens showed a pseudo-ductile behavior with progressive failure regardless of the quantity of carbon fiber or of the type of high elongation fiber. Bakis et al. also noted that a good pseudo-ductile behavior was obtained for the specimen with well-dispersed carbon fiber.

Recently, Yujin et al. [7] performed tensile tests on hybrid rods with carbon and glass fibers. Test results revealed that hybrid rods with carbon fiber showed a distinct pseudo-ductile behavior where a number of subsequent stress peak are repeated. Yujin et al. also observed that, for specimens with carbon fiber concentrated in the core, a secondary failure occurred at stress larger than the maximum stress developed at the initial failure.

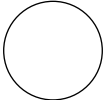
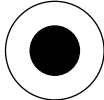
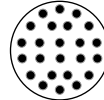
3. Experiments

This study considers glass and carbon fibers in order to investigate a hybrid effect according to the mixing of different types of fibers. Five types of FRP rod samples were fabricated as summarized in Table 1. Vinyl-ester (VE) and unsaturated polyester (PE) were used as resins. A total of 40 samples consisting of 10 four-sample group for each design parameter were fabricated and tested. For the design of the hybrid FRP samples, the tensile strength of the rods was assumed to be 900 MPa and the volume fraction of the fibers was calculated based on the commonly known values for the average tensile strengths of glass and carbon fibers.

Since the bond performance between the FRP rebar and concrete was beyond the scope of this study, the hybrid rods were fabricated with a circular cross-section of diameter of 12.7 mm. E-glass fiber (RS-4400-366, Owens Corning Korea) and carbon fiber (T-700 12K, Toray Inc.) were used in this study. Vinyl-ester and unsaturated polyester are known as effective resins for the pultrusion process of fabrication because they offer economical efficiency, low viscosity, and rapid hardening. The material properties of fibers and resins used in this study are provided in Table 2.

The tensile tests of the FRP rods were carried out in accordance with ASTM D 3916 [8]. The length of the

Table 1
Type of FRP rod samples

Type	A	B	C	D	E
Fiber volume fraction by weight (%)	Glass: 49	Carbon: 39	Glass: 37 Carbon: 23	Glass: 37 Carbon: 23	Glass: 37 Carbon: 23
Carbon fiber placement	None	Full	In core	On surface	Dispersed
Cross-section ^a					

^a Color key: white (glass fiber), black (carbon fiber).

Table 2
Material properties of fibers and resins

Material	Tensile strength (MPa)	Elastic modulus (GPa)	Elongation (%)
Carbon fiber	4900	230.0	2.10
E-glass fiber	2410	79.0	3.04
Vinylester resin	79	3.7	7.00
Unsaturated polyester resin	62	3.1	5.00

specimens was 850 mm and an UTM with a capacity of 1000 kN was used. Strains were measured with an extensometer and a strain gauge mounted at the center of the specimens. The specimens were fixed at both the top and the bottom with a pair of aluminum grip adapters.

4. Results and discussion

The tensile strength of the specimen was calculated by dividing the measured maximum load by the cross-sectional area of the rod (A_{rod}). The elastic modulus of the FRP rod (E_{rod}) was calculated using the following expression as recommended in the regulations of CSA S806-02 [9].

$$E_{rod} = \frac{(P_1 - P_2)}{(\varepsilon_1 - \varepsilon_2)A_{rod}} \quad (3)$$

where, P_1 and P_2 are the applied loads corresponding to 50% and 25% of the ultimate load respectively, and ε_1 and ε_2 are the corresponding strains.

4.1. Tensile properties of glass/carbon rods

Table 3 summarizes the mechanical properties of the non-hybrid specimens with all glass and carbon fibers

(Types A and B). The stress-strain relationship of the specimens exhibited the perfectly linear behavior with the occurrence of brittle fracture. Most of the specimens failed at the gauge length, but some of them presented ruptures at the grip adapters. The average of experimental results was 21% higher than the predicted ones for the elastic modulus and 24% lower for the tensile strength.

According to the test data in the literatures, strength reduction may occur due to damage, misplacement of fibers during the fabrication, the size of specimen, and the gripping method. The ultimate strain developed in carbon fiber composite generally produces a coupon test value 15–20% lower than the non-impregnated strand. In addition, the tests performed by Malvar and Bish [10] on different grips like ASTM D 3916 grip adapter and clamp type revealed that the tensile strength of FRP rod obtained using ASTM D 3916 grip adapters is the lowest. Moreover, Kretsis [3] stated that the values predicted by Eq. (1) exhibit large standard deviation because this expression is a linear equation on the assumption that resin is transferring perfectly loads between fibers.

Regard to this effect stated above, the experimental and predicted values may show difference to some extent. However, as provided in Table 3, the experimental values were consistent since the coefficients of variation were relatively small. The difference between experimental values and predicted ones will not be discussed in detail as this study focuses on the examination of the relative behavioral changes according to the arrangement of fibers within the section of the rod.

On the other hand, when the experimental results obtained for the specimens using unsaturated polyester resin is compared with the vinylester resin, a difference of

Table 3
Results of tensile tests for non-hybrid rod specimens

Type	Experiment					Prediction ^a		
	Tensile strength		Elastic modulus		Ultimate strain ^b	Tensile strength	Elastic modulus	Ultimate strain
	(MPa)	COV (%)	(MPa)	COV (%)	(%)	(MPa)	(MPa)	(%)
A-VE	972	2.3	47.8	3.7	2.03	1248	41.1	3.04
A-PE	983	5.4	48.3	0.8	2.03	1240	40.8	3.04
B-VE	1454	10.6	114.8	4.3	1.27	1935	92.1	2.10
B-PE	1431	4.0	114.6	3.2	1.25	1928	91.8	2.10

^a Calculated value using Eqs. (1) and (2).

^b Calculated value using test data and COV = coefficient of variation.

Table 4
Results of tensile tests for hybrid rod specimens

Type	Tensile strength ^a		Elastic modulus ^b		Ultimate strain ^c	Hybrid effect ^d
	(MPa)	COV (%)	(GPa)	COV (%)	(%)	(%)
C-VE	1281	5.4	80.4	3.1	1.59	26
C-PE	1331	3.2	83.1	4.2	1.60	27
D-VE	1083	3.2	78.9	0.5	1.37	9
D-PE	1128	2.7	79.5	0.4	1.42	13
E-VE	1045	2.6	62.4	0.4	1.68	33
E-PE	1213	4.9	84.0	2.3	1.44	14

Notes: $c = a/b$, $d = c/\epsilon_t^{\text{ult}}$ of non-hybrid carbon rod.

1.6% occurred. Therefore it may be concluded that the variations of the elastic modulus and tensile strength are insensitive to the type of resin.

4.2. Hybrid effects of hybrid rods

Table 4 summarizes the results of tensile tests for the hybrid rod specimens. The hybrid effect is expressed in reference to the ultimate strain of the non-hybrid carbon FRP rod.

For the hybrid rods composed of 37% of glass fiber and 23% of carbon fiber (Types C, D, and E), a hybrid effect appeared as the ultimate strain increases unlike that of non-hybrid carbon rod. For the rods with vinylester resin, the highest hybrid effect was developed in Specimen E-VE with carbon fiber dispersed over the section of the rod and the lowest in Specimen D-VE with carbon fiber arranged on the surface of the rod. As the rod with unsaturated polyester resin showed relatively smaller hybrid effect, the reason may be related to the arrangement of the carbon fiber.

Overall, the ultimate strains of the hybrid rods increased 9–33% compared to that of the non-hybrid carbon rod. However, the elastic modulus and tensile strength of the hybrid rods according to the arrangement method of carbon fiber were not be proportional to the hybrid effect, as provided in Table 4. Specimen E-VE shows the highest hybrid effect for the rod with vinylester resin, while Specimen C-VE shows the highest elastic modulus and tensile strength. Therefore, if hybrid rods are applied to the structures, they should be designed in a way to develop the tensile characteristics fitting for the design purpose.

4.3. Effects of fiber arrangement in hybrid rods

Since the roving used in the fabrication of rods samples was a bundle of fibers, it was difficult to obtain the desired arrangement during the fabrication process of hybrid rod. In addition, when the straightness of roving is not maintained during the pultrusion process, the fibers may not be arranged uniformly. Fig. 1 presents the cross-section of the samples at two different locations and illustrates that they cannot correctly arranged along the length of the samples as desired.

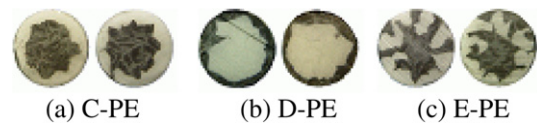


Fig. 1. Cross-sections of rod specimens at different location.

As provided in Table 4, regardless of the type of resin, Specimens C-VE and C-PE with the carbon fiber disposed in the core of the rod exhibit tensile strength and ultimate strain larger by about 18% and 14% than Specimens D-VE and D-PE with the carbon fiber disposed on the surface. This shows that it is better to arrange carbon fiber in the core of the rod in view of the tensile strength.

The tensile strength of Specimens E-VE and E-PE with the dispersed carbon fiber was reduced 18% for vinylester resin and 9% for unsaturated polyester resin, compared to the Specimen Type C with the carbon fiber arranged in the core, while the ultimate strain increased 6% for vinylester resin and decreased 10% for unsaturated polyester resin. The variation observed according to the types of resin was nearly independent of the effect of resin, as previously explained, which may be due to the difference in the dispersion. As the cross-sections of specimens shown in Fig. 2(c) and (f), the carbon fiber of Specimen E-PE was relatively more concentrated in the core than Specimen E-VE, Specimens E-PE showed a behavior similar to Specimen C-PE.

4.4. Tensile behaviors of hybrid rods

For hybrid rods with vinylester resin, the stress–strain curve behaves linearly until the ultimate stress and shows the stress redistribution beyond this stress level, as shown in Fig. 2(a)–(c). On the other hand, such behavior could not be clearly observed in hybrid rods with the unsaturated polyester resin. These results may result from the differences in the characteristics and performance of the resin redistributing loads between the fibers.

Bakis et al. [6] stated that a distinct pseudo-ductile behavior was observed for specimens with dispersed carbon fiber. However, as shown in Fig. 2(c) and (f), the behavior of the specimens fabricated in this study exhibited few repetition of subsequent stress peak similar to the experiment results by Yujin et al. [7]. In addition, Yujin

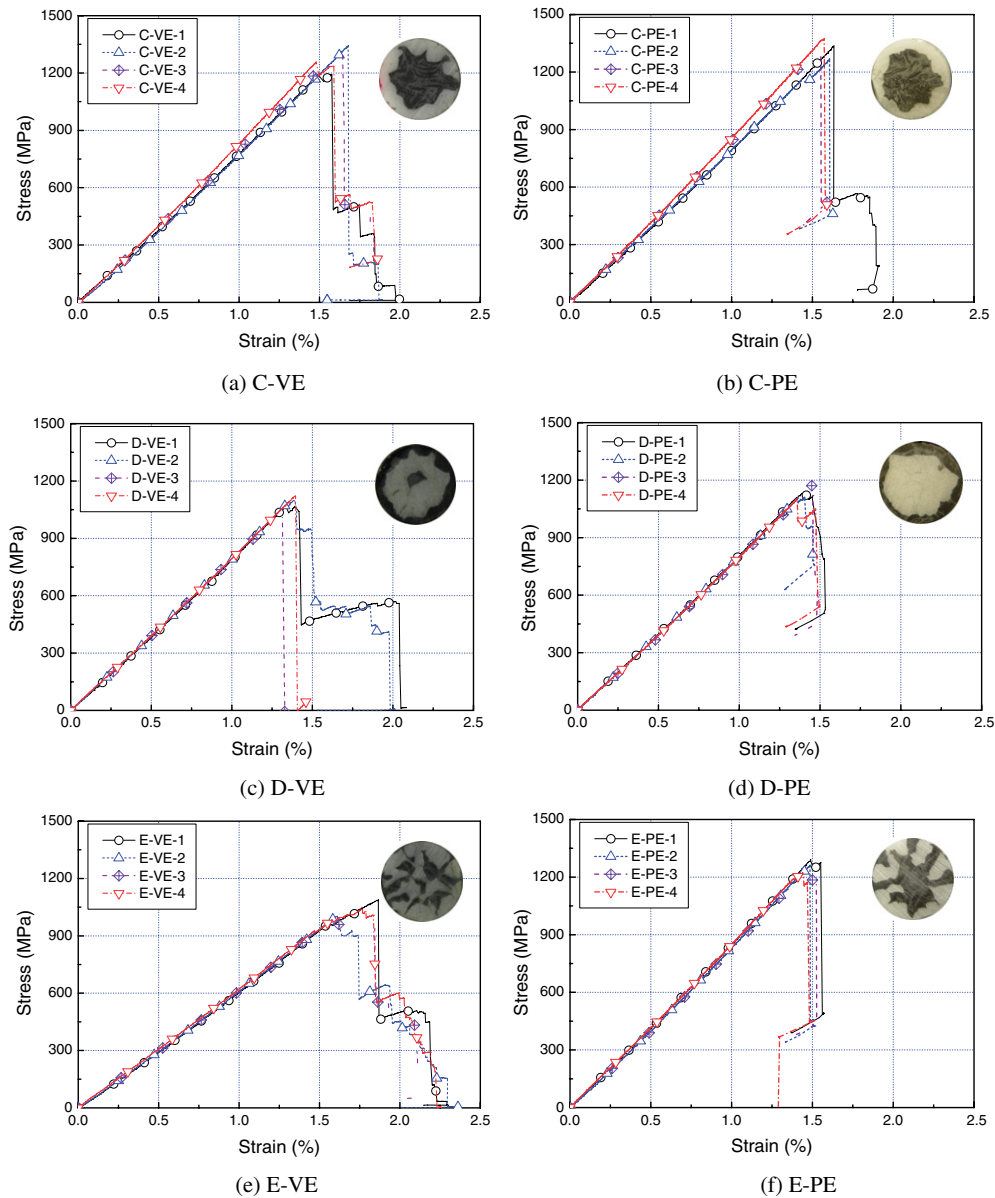


Fig. 2. Stress versus strain curves for hybrid rod specimens.

et al. reported that, when carbon fiber is arranged on the surface, a secondary failure occurs at the stress larger than the one developed at the initial failure. However, this behavior was not observed in the specimens tested in this study.

Such differences stated above may be attributed to the different volume fractions of carbon and glass fibers used by the researchers. Jones and DiBenedetto [4] reported that low elongation fiber should be well dispersed in small quantity to exhibit a pseudo-ductile behavior. Though the ratios of carbon fiber to glass fiber used by Yujin et al. [7] and Bakis et al. [6] were 27% and 37%, respectively, a relatively larger value of 61% was assumed in this study. Moreover, when the fibers are dispersed randomly, the stress from carbon fiber to glass fiber does not satisfactorily redistributed resulting in the release of the energy.

The energy is produced by the failure of carbon fiber which, in turn, affects the neighboring glass fiber and, subsequently, leads to the trends diverging from the predictions [7].

5. Summary and conclusions

The results of test program on the hybrid FRP rods according to the type of resin and the arrangement method of fibers can be summarized as follows.

The ultimate strains of the hybrid rods composed of 37% of glass fiber and 23% of carbon fiber were seen to increase 3–33% compared to the non-hybrid carbon FRP rod. The hybrid effect was obtained by the mixing of fibers.

The ultimate strain of Specimen E-VE with carbon fiber dispersed in the section of the rod increased 33% compared

to Specimen B-VE composed only of carbon fiber and resin. However, the increase rate of the ultimate strain may vary as observed in Specimen E-PE as the dispersion of carbon fiber in the section can change if the continuity and precision of fabrication could not be secured.

Differences in the tensile behavior occurred according to the method adopted to arrange a given quantity of glass and carbon fibers. Regardless of the type of resin, the tensile strength, elastic modulus, and ultimate strain of the specimens (Type C) with carbon fiber arranged in the core were seen to increase more compared to the Specimens Type D with the carbon fiber arranged on the surface. The tensile strength of Specimen E-VE with the carbon fiber dispersed over the section was smaller than that of Specimen C-VE with the carbon fiber concentrated in the core, while its ultimate strain of the one was larger than the other. This trend did not appear in the specimens with unsaturated polyester resin, which is because the carbon fiber was relatively more concentrated in the core during the fabrication process.

The hybrid effect was the largest where carbon fiber was well dispersed in the section, while the tensile strength was largest where carbon fiber was concentrated in the core. Since the extent of increase of hybrid effect, tensile strength, and elastic modulus varies according to the arrangement methods of the carbon fiber, it should be designed in a way to develop tensile characteristics fitting for the design objective when hybrid rods are applied to the structures. Given the difficulty of fabrication and the increase of the ultimate strain, elastic modulus, and tensile strength as a whole, the carbon fiber had better be arranged in the core.

The distinct pseudo-ductile behavior of hybrid rods was not observed in this study mainly because the volume fraction of carbon fiber was more used than that of glass fiber. Further investigation should be conducted to study the

effect of the stress redistribution mechanism on the pseudo-ductile behavior regard to the quantity as well as the dispersion of carbon fiber.

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